



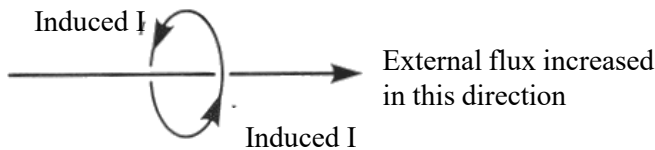
Can you produce electricity without a battery or create an electric field without any charges?

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Week 10/11
Serway Chap 30

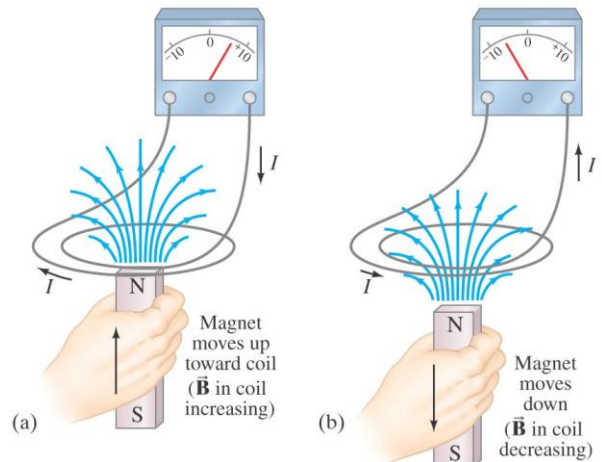
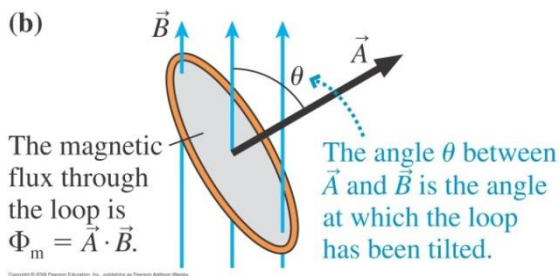
"I was at first almost frightened when I saw such mathematical force made to bear upon the subject, and then wondered to see that the subject stood it so well."

- Faraday's letter to James Clerk Maxwell (25 March 1857)



$$\text{Induced e. m. f.}, \mathcal{E} = -N \frac{d\Phi_B}{dt}$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

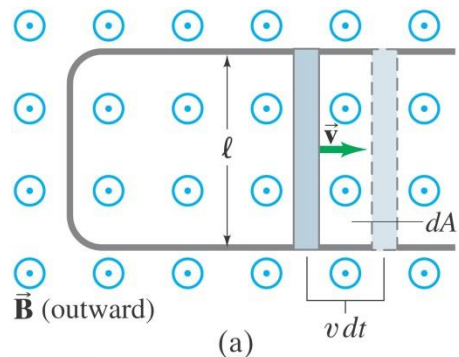


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[Knight pg 1050, Fig 34.16]

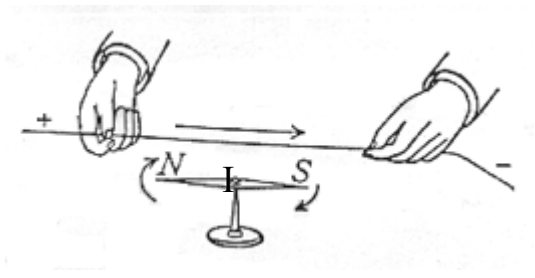
Self-check – after completing the lessons (lectures and tutorials) for this topic, are you able to

1. apply Faraday's law and Lenz's law to determine induced e.m.f. / current
2. integrate the concepts in magnetic deflection forces, sources of magnetic fields and electromagnetic induction above to solve problems in electromagnetism

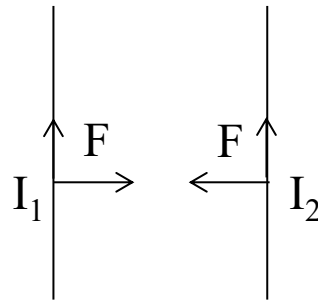


$$\text{Induced e. m. f.}, \mathcal{E} = Blv \cos \theta$$

Outline of historical development of Electromagnetism



Oersted, 1820

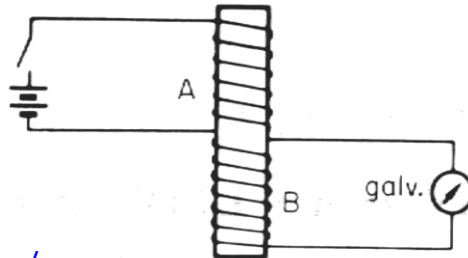


Ampere, 1822

After Ampere and others investigated the magnetic effects of a current, Faraday tried to find the reverse. He tried to produce a current by using a magnetic field. He began work in 1825 and it took him six years before he found something conclusive. Below is a diagram of the setup up that led to this very important discovery.



https://www.youtube.com/watch?v=jeTmla00_rc&t=5s



Faraday, 1831



<https://www.youtube.com/watch?v=txmKr69jGBk&t=20s>

Defining magnetic flux

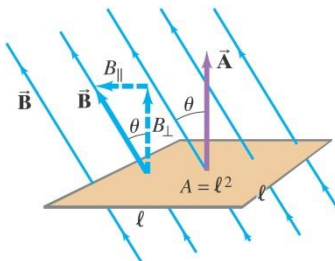
In a similar manner as we have defined electric flux earlier, we will now define magnetic flux

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

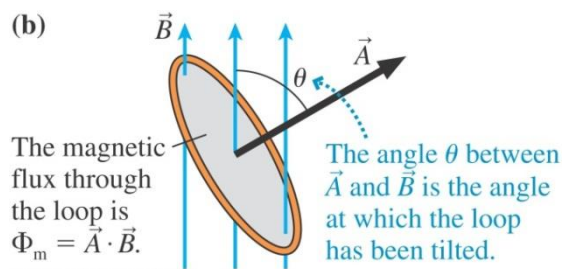
through an area of loop. The unit for magnetic flux is Weber (Wb).

$$1\text{Wb} = 1 \text{ T m}^2$$

The flux through each elemental area is given by the dot product $\vec{B} \cdot d\vec{A} = |\vec{B}||d\vec{A}| \cos \theta$ as illustrated below



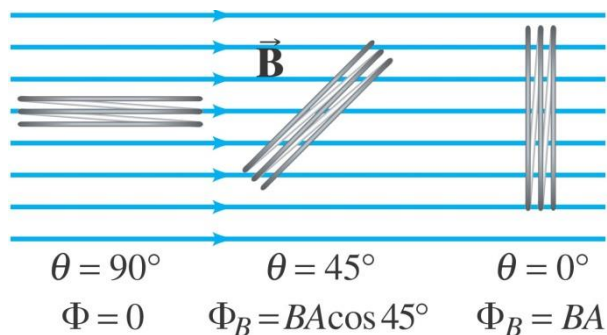
[Giancoli pg 760, Fig 29.3]



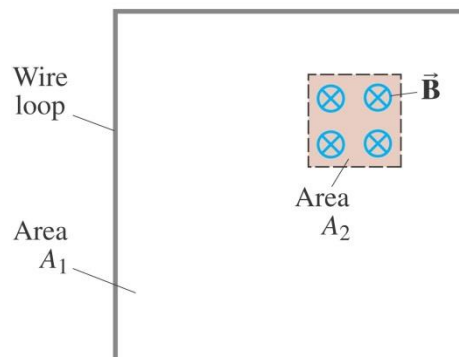
[Knight pg 1050, Fig 34.16]

Magnetic flux

The magnetic flux is analogous to the electric flux – it is proportional to the total number of magnetic field lines passing through the loop.

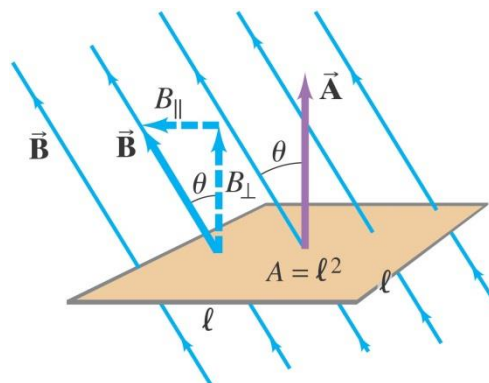


[Giancoli pg 760, Fig 29.4, Fig 29.5]



$$\Phi_B = BA_2, \text{ and not } BA_1$$

Example G29-2 (modified): A loop of wire in a magnetic field. A square loop of wire of side $l = 5.0$ cm is in a uniform magnetic field $B = 0.16$ T. What is the magnetic flux in the loop (a) when B is perpendicular to the face of the loop and (b) when B is at an angle of 30° to the area A of the loop? (c) What is the change in magnetic flux through the loop when it rotates from position in (b) to position in (a)?



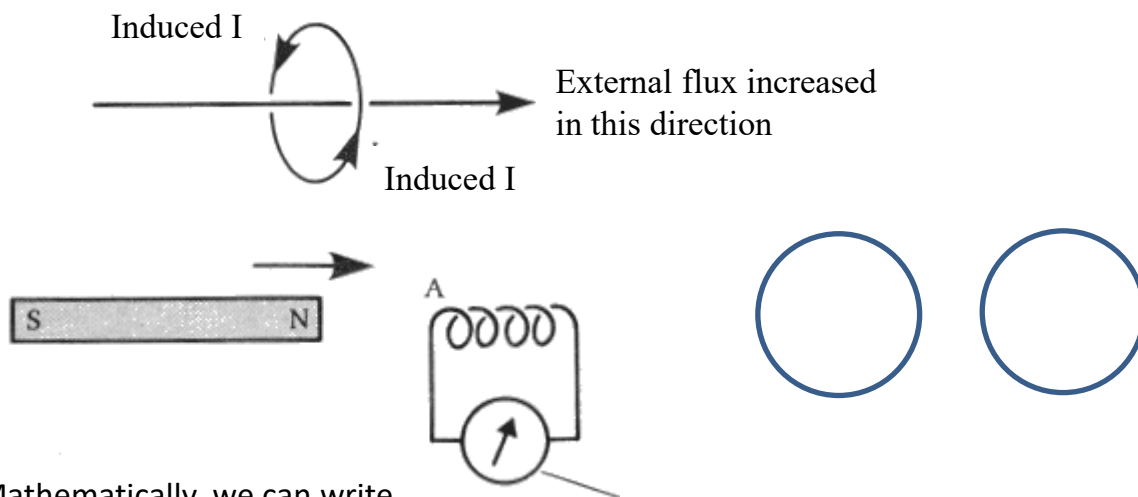
Magnetic flux linkage

Magnetic flux linkage is the product of magnetic flux and the number of turns (N) of coil: $N\Phi_B$. We can now define Faraday's Law.

Faraday's Law and Lenz's Law

Faraday's Law states that the magnitude of the induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage or to the rate of 'cutting' of magnetic flux.

Lenz's Law states that the direction of any induced emf is such as to oppose the change in the flux that causes it. See example below,



Mathematically, we can write

where

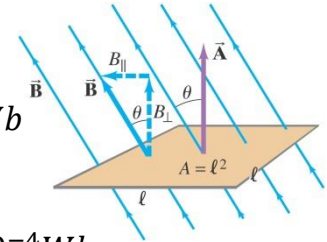
$$\text{Induced e. m. f.}, \mathcal{E} = -N \frac{d\Phi_B}{dt}$$

Pointers to note:

$$\Phi_B = \int \vec{B} \cdot d\vec{A} (= BA \text{ for uniform } B).$$

1. Faraday's Law refers to induced e.m.f. (applicable to both open and closed circuits)
2. Lenz's Law refers to induced current. (applicable only to closed circuits. If the circuit is open, we can usually think in terms of what would happen if it were closed and in this way find the direction of the induced e.m.f.)
3. The -ve sign in mathematical form of law of the electromagnetism is due to Lenz's Law.
4. From the conservation of energy, the work done against resistive forces due to induced current is always equal to the internal energy produced in the coil.

Example G29-2: A loop of wire in a magnetic field. A square loop of wire of side $l = 5.0 \text{ cm}$ is in a uniform magnetic field $B = 0.16 \text{ T}$. What is the magnetic flux in the loop (a) when B is perpendicular to the face of the loop and (b) when B is at an angle of 30° to the area A of the loop? (c) What is the change in magnetic flux through the loop when rotates from position in (b) to position in (a)? (d) What is the magnitude of the average current in the loop if it has a resistance of 0.012Ω and it is rotated from position (b) to position (a) in 0.14 s ?



- a) $\Phi_{Bf} = \vec{B} \cdot \vec{A} = BA \cos 0^\circ$
 $= (0.16 \text{ T})(2.5 \times 10^{-3} \text{ m}^2)(1) = 4.0 \times 10^{-4} \text{ Wb}$
- b) $\Phi_{Bi} = \vec{B} \cdot \vec{A} = BA \cos 30^\circ$
 $= (0.16 \text{ T})(2.5 \times 10^{-3} \text{ m}^2)(\cos 30^\circ) = 3.5 \times 10^{-4} \text{ Wb}$
- c) $\Delta\Phi_B = \Phi_{Bf} - \Phi_{Bi} = 0.5 \times 10^{-4} \text{ Wb}$
- d)

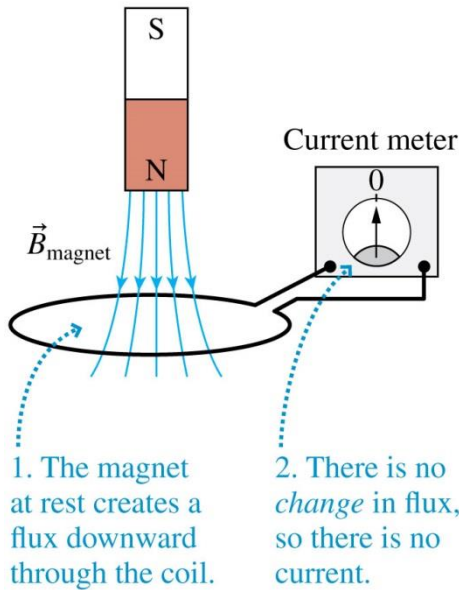
Example: Flux through loop [Giancoli 779, Prob 29.18]

The magnetic flux through each loop of a 75-loop coil is given by

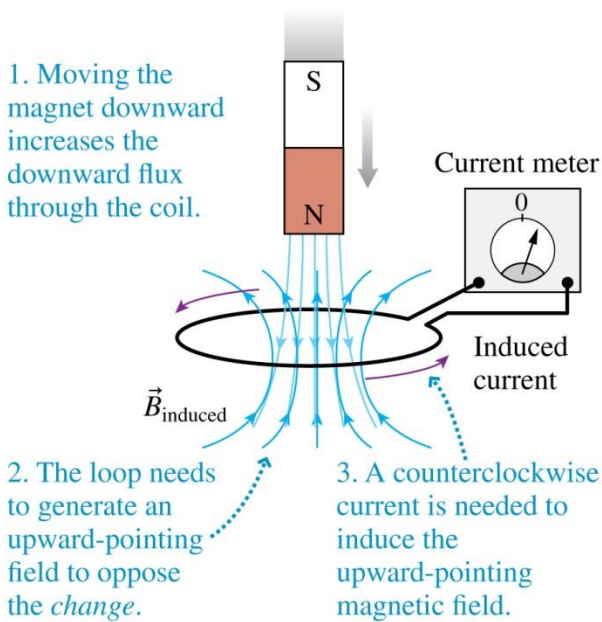
$((8.8t - 0.51t^3) \times 10^{-2}) \text{ T m}^2$, where the time t is in seconds. (a) Determine the emf $\mathcal{E}$ as a function of time. (b) What is $\mathcal{E}$ at $t = 1.0 \text{ s}$ and $t = 4.0 \text{ s}$?

Lenz's Law (Knight Chap 25)

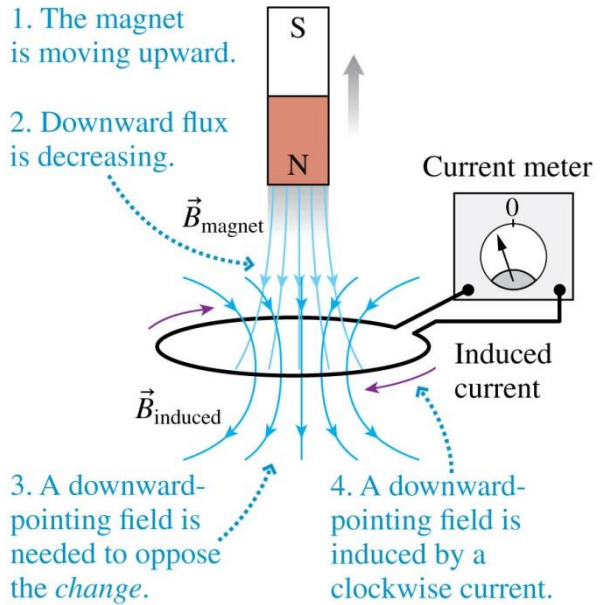
(a) Magnet at rest



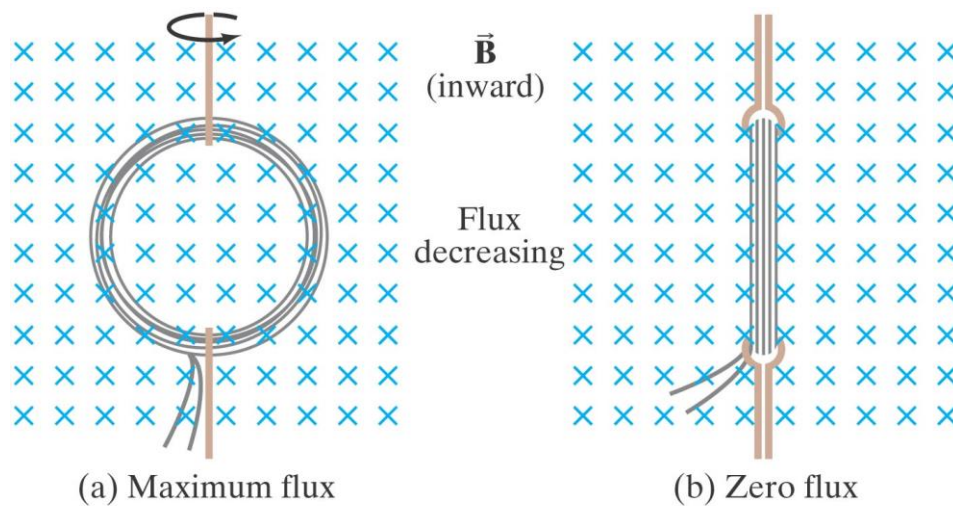
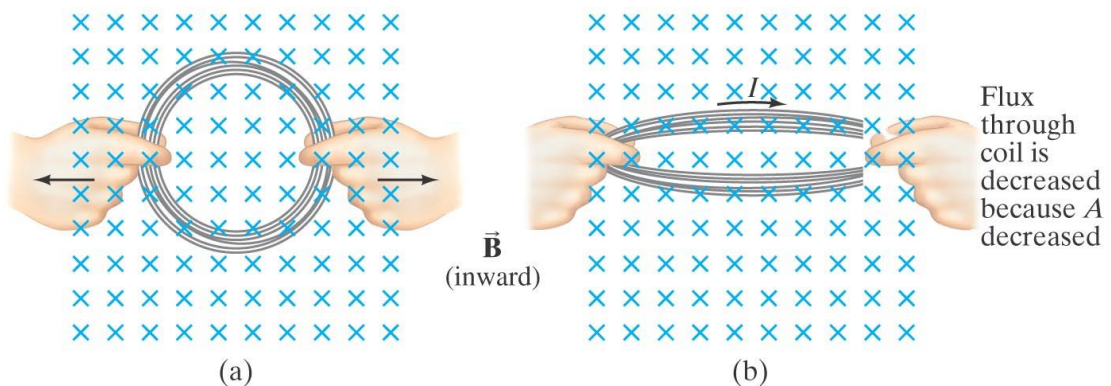
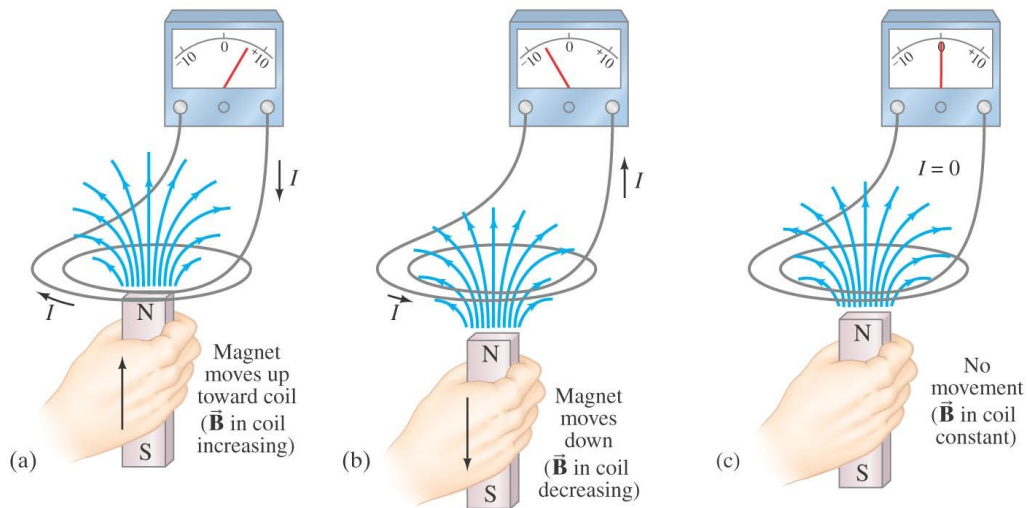
(b) Magnet moving down



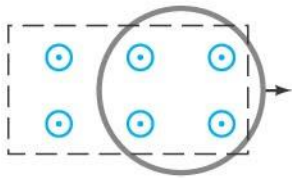
(c) Magnet moving up



Some ways of inducing e.m.f. [Giancoli pg 759, 762]

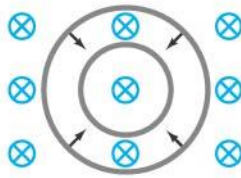


Direction of Induced e.m.f. [Giancoli pg 763, 764]



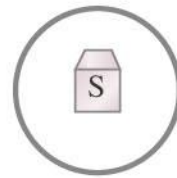
(a)

Pulling a round loop to the right out of a magnetic field which points out of the page



(b)

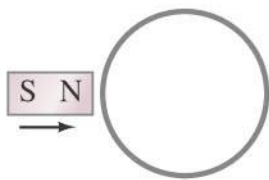
Shrinking a loop in a magnetic field pointing into the page



(c)

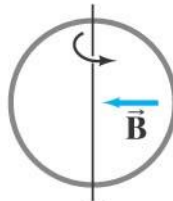
S magnetic pole moving from below, up toward the loop

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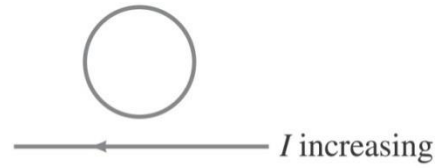
(d)

N magnetic pole moving toward loop in the plane of the page



(e)

Rotating the loop by pulling the left side toward us and pushing the right side in; the magnetic field points from right to left



(f)



(a)



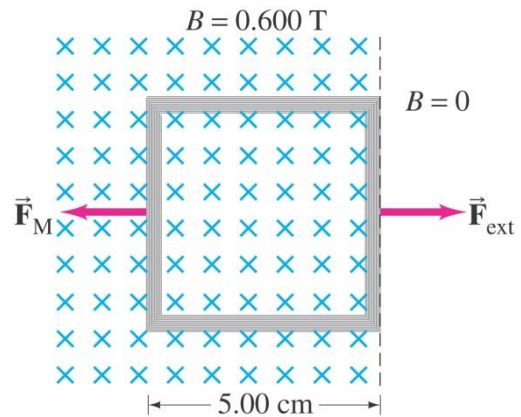
(b)

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[Giancoli pg 789, Fig 30.4]

Giancoli pg 764 Example 29.5

A 100-loop square coil of wire, with side $l = 5.00 \text{ cm}$ and total resistance 100Ω , is positioned perpendicular to a uniform 0.600-T magnetic field. It is quickly pulled from the field at constant speed (moving perpendicular to B) to a region where B drops abruptly to zero. At $t = 0$, the right edge of the coil is at the edge of the field. It takes 0.100 s for the whole coil to reach the field-free region. Find (a) the rate of change in flux through the coil, and (b) the emf and current induced. (c) How much energy is dissipated in the coil? (d) What was the average force required (F_{ext})?



- (a) We want to find out how the magnetic flux $\Phi_B = BA$ changes in the time interval $\Delta t = 0.1 \text{ s}$.

$$\frac{\Delta \Phi_B}{\Delta t} = \frac{0 - (0.60 \text{ T}) \times (0.05)^2}{0.01} = -1.50 \times 10^{-2} \text{ Wb/s}$$

- (b) Induced e.m.f. $\mathcal{E} = -N \frac{\Delta \Phi_B}{\Delta t} = -100 \times (-1.50 \times 10^{-2}) = 1.50 \text{ V}$

By Ohm's law, $I = \frac{\mathcal{E}}{R} = \frac{1.50 \text{ V}}{100 \Omega} = 1.50 \times 10^{-2} \text{ A}$ (clockwise)

- (c) Total energy $E = Pt = I^2 R t = (1.50 \times 10^{-2})^2 \times (100 \Omega)(0.100 \text{ s}) = 2.25 \times 10^{-3} \text{ J}$

(d) $F = \frac{W}{d} = \frac{2.25 \times 10^{-3} \text{ J}}{5.00 \times 10^{-2} \text{ m}} = 0.0450 \text{ N}$

Or

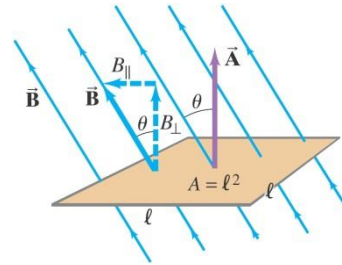
$$F = NBIL = 100 \times 0.0150 \text{ A} \times 0.0500 \text{ m} \times 0.600 \text{ T} = 0.045 \text{ N}$$

Example: Electric Generators [Giancoli 766, Fig 29.15]

A generator is the opposite of a motor – it transforms mechanical energy into electrical energy. This is an ac generator. The axle is rotated by an external force such as falling water or steam. The brushes are in constant electrical contact with the slip rings.

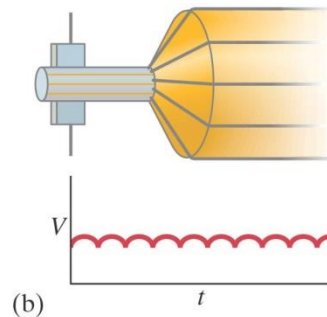
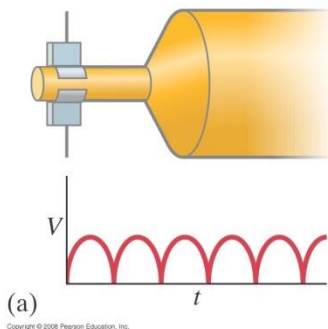
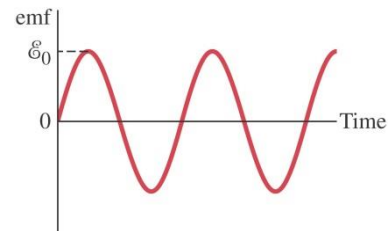
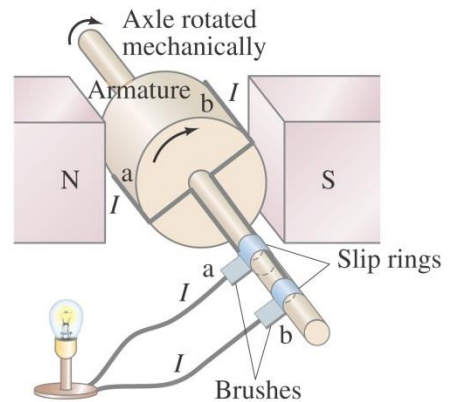
$$\Phi_B = \int \vec{B} \cdot d\vec{A} = BA \cos \theta$$

If the coil is rotating at constant angular velocity such that $\theta = \omega t$, taking $\theta_0 = 0$ at $t = 0$.



The induced e.m.f in the coil

$$\mathcal{E} = -\frac{d\Phi_B}{dt} =$$



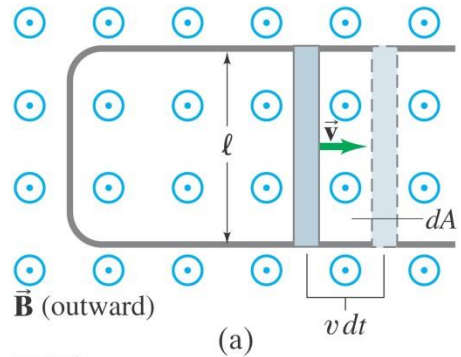
Giancoli Fig 29-18 (a) A dc generator with one set of commutator, and (b) a dc generator with many sets of commutators and windings.

EMF induced in a moving conductor [Giancoli pg 765]

What will happen if we move a straight conductor of length l at constant speed v_x perpendicularly through a magnetic field $\vec{B}$ (directed into the page)? Now, we have no coil, how do we apply Faraday's law?

We note here that at time dt , the area of loop increases by $dA = ldx = lv_x dt$. Thus, by Faraday's law, the magnitude of the induced emf

$$\mathcal{E} = \frac{d\Phi_B}{dt} = B \frac{dA}{dt} =$$



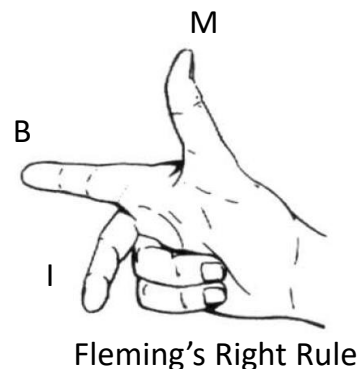
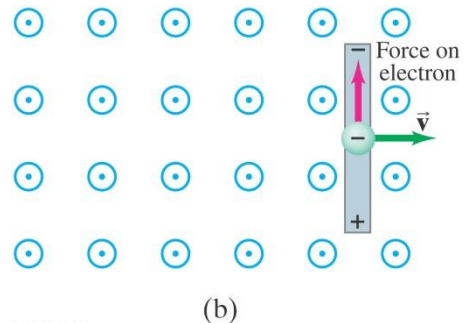
In fact, we do not need U-shape conductor, there will be an induced emf in the conducting rod with it. Consider the microscopic picture where the electrons in the in the conductor experience a force $\vec{F}_B = q \vec{v} \times \vec{B}$ that is directed along the length l , perpendicular $\vec{v}$ and $\vec{B}$. Under the influence of this force, the electrons move away the lower end of the conductor and leave a positive charge at the lower end. Thus, an electric field $\vec{E}$ is produced inside the conductor. In equilibrium, the upward magnetic force qvB is balanced by the qE ,

$$qvB = qE \text{ or } E = vB.$$

The potential difference between the ends of the conductor is given by $\Delta V = El$. Thus, we get

$$\Delta V = Blv.$$

Thus, a potential difference is set up at the ends of the rod due to the separation of charges.



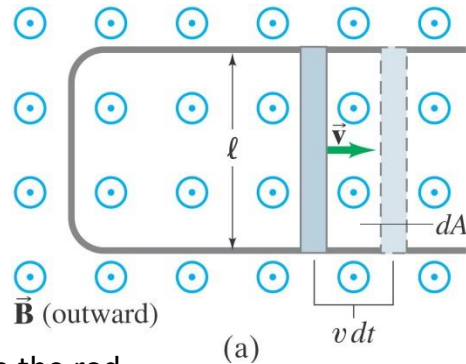
Force on Rod [Modified Giancoli Example 29.8 pg 766]

To make the rod of length l move to the right at speed v , you need to apply an external force F on the rod to the right. The resistance of the rod is R and the magnetic field is B . (a) Explain and determine the magnitude of the required force. (b) What external power is needed to move the rod?

Induced current $I = \frac{\mathcal{E}}{R} = Blv \sin 90^\circ$

a)

$$F = BIl \sin 90^\circ$$

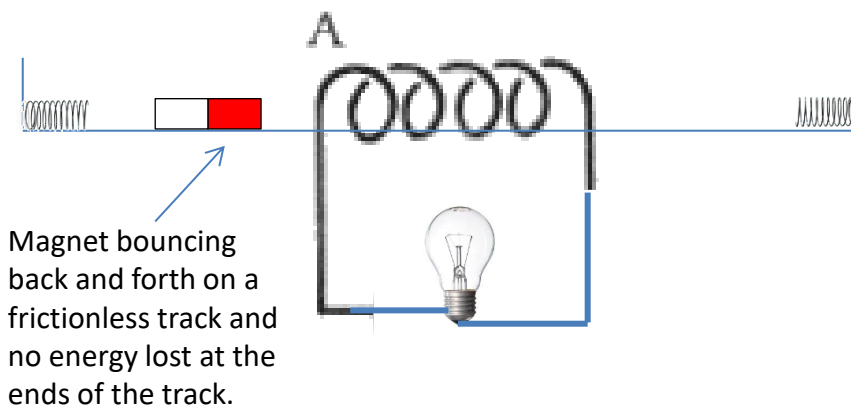


b) External mechanical power needed to move the rod

$$P_{mech} = Fv =$$

Electrical power dissipated in resistance $P_{ele} = I^2 R =$

Question to ponder: Will the bulb light up forever? Free energy?



Back emf (Giancoli Chap 29.5)

(Giancoli page 768) When the armature of a motor turns, the magnetic flux through the coil changes and an emf is generated. This emf acts to oppose the motion (Lenz's Law) and is called the **back emf or counter emf**. The greater the speed of the motor, the greater the back emf. A motor normally turns and does work. However, if there is no load, the motor's speed will increase until the back emf equals to the input voltage. When there is a load, the speed may be limited by the load. The back emf will be less than the external applied voltage.

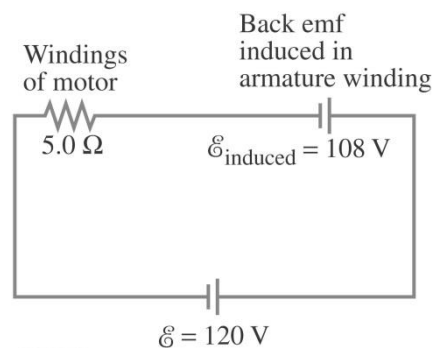
Example: The induced current in a motor [Giancoli, Example 29.10, pg 769]

The armature windings of a dc motor have a resistance of $5.0\ \Omega$. The motor is connected to a 120-V line, and when the motor reaches full speed against its normal load, the back emf is 108 V. Calculate (a) the current into the motor when it is just starting up, and (b) the current when the motor reaches full speed.

a) To start up, $I = \frac{V}{R} = \frac{120\text{V}}{5.0\Omega} = 24\text{ A}$

b) $120\text{ V} - 108\text{ V} = I (5.0\ \Omega).$

$$I = \frac{12\text{ V}}{5.0\ \Omega} = 2.4\text{ A}$$

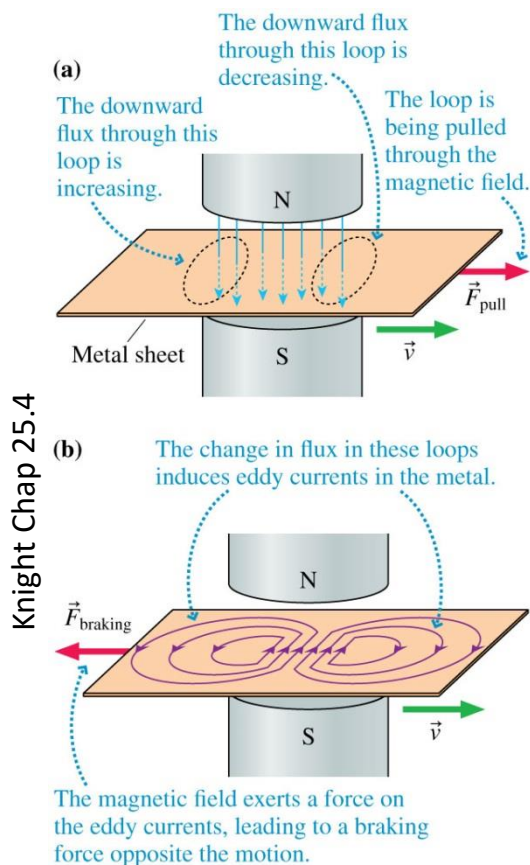
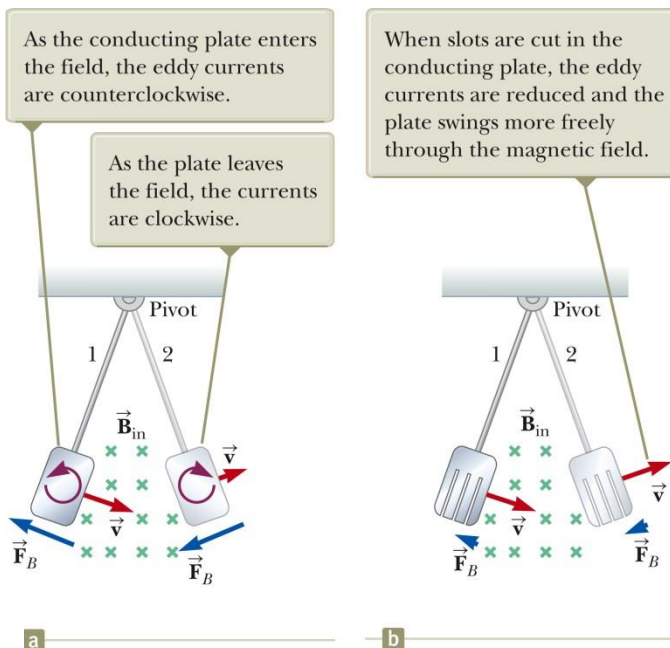


Eddy Currents (Giancoli Chap 29.5) EEE Video after Exam

Induced currents are not always confined to well-defined path in wires. They can also flow in bulk material. These are called eddy currents, and can dramatically slow a conductor moving into or out of a magnetic field.

[Serway pg 911, Fig 31.22]

An e.m.f. and current can be induced when there is a changing magnetic flux. Similarly, circulating currents called eddy currents are induced in bulk pieces of metal moving through a magnetic field. A retarding force is produced when the conducting plate enters or leaves the magnetic field. This force can be reduced by cutting slots in the plate.



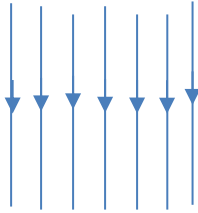
There are two “loops” lying entirely in a metal sheet between two magnets. As the sheet is pulled, the loop on the right is leaving the magnetic field, and the flux is decreasing. According to Faraday’s law, the flux change induces a current to flow around the loop. Lenz’s law says the current flows clockwise.

Both whirlpools of current are moving in the same direction as they pass through the magnet. The magnetic field exerts a force on the current, opposite the direction of pull, acting as a *braking force*.

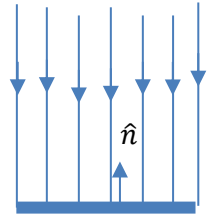
Force, Field and Flux



Force, $\vec{F}$



Field Lines, $\vec{E}$

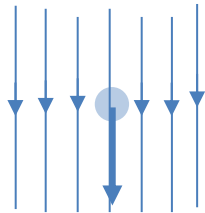


Area A

Flux Φ_E

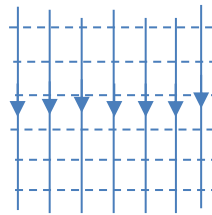
$$\vec{F} = q\vec{E}$$

$$\Phi_E = \int \vec{E} \cdot d\hat{A}$$



Force + Field

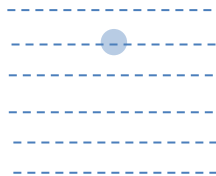
$$\vec{E} = -\frac{dV}{dr}$$



Equipotential Lines V

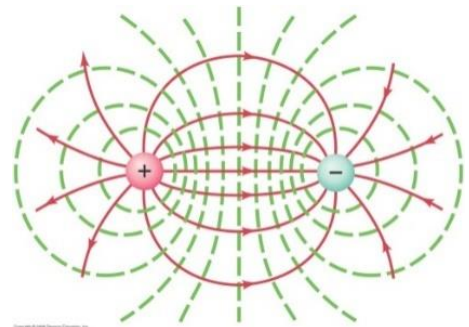
$$\vec{F} = -\frac{dU}{dr}$$

$$U = qV$$



Potential Energy

Using dipole as example



A more abstract treatment of Fields and Derivations of Expressions for Electric Fields and Magnetic Fields

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. apply Gauss's law to analyze the relationship between charge and electric fields
2. apply Ampere's law to analyze the relationship between current and magnetic fields



2026S1 PH1012: Physics A
Gauss's Law and Ampere's Law

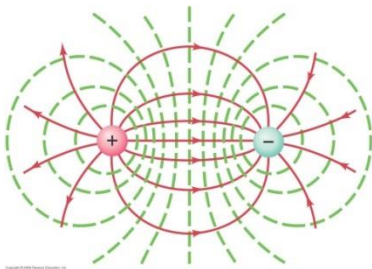
Dr Ho Shen Yong
Lecturer, School of Physical and Mathematical Sciences
Nanyang Technological University

Week 11
Serway 23

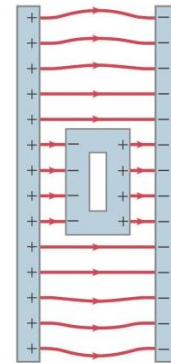
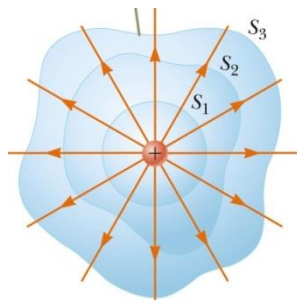
"Genius is one percent inspiration, ninety-nine percent perspiration."

- Thomas Alva Edison (1847-1931)

E-field / potential of
system of charges



Gauss's Law



Electrostatic
equilibrium

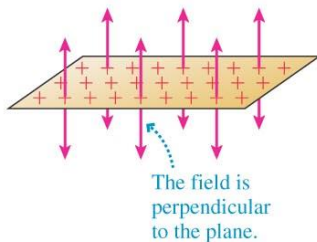


<https://www.youtube.com/watch?v=pqTfk9HMLj4&t=2s>

Three key charge
distributions related
charge storing
configurations

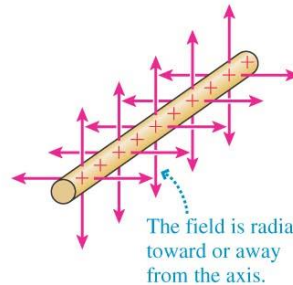
Basic
symmetry:

Planar symmetry



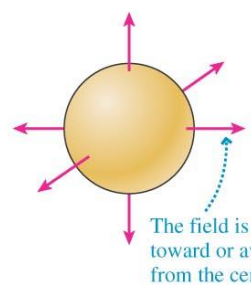
The field is
perpendicular
to the plane.

Cylindrical symmetry



The field is radial
toward or away
from the axis.

Spherical symmetry



The field is radial
toward or away
from the center.

More
complex
example:



Infinite parallel-plate capacitor



Coaxial cylinders

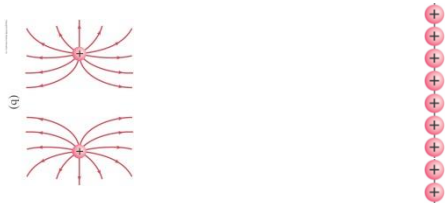


Concentric spheres

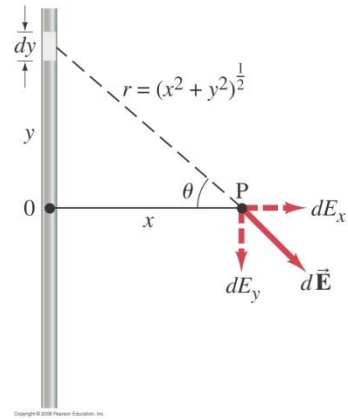
Electric Fields

Giancoli pg 573, Example 21.11:

Electric field due to a long line of charge



Determine the magnitude of the electric field at any point P a distance x from a very long line (a wire, say) of uniformly distributed charge. Assume x is much smaller than the length of the wire, and let λ be the charge per unit length (C/m).



$$dE = \frac{1}{4\pi\epsilon_0} \frac{dQ}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dy}{(x^2 + y^2)}$$

P is along bisector of rod so y component of $\vec{E}$ is zero.

$$E_y = \int dE \sin \theta = 0$$

Along the x direction,

$$E_x = \int dE \cos \theta = \frac{\lambda}{4\pi\epsilon_0} \int \frac{\cos \theta dy}{(x^2 + y^2)} = \dots$$

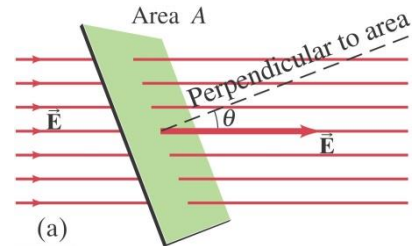
For an infinitely long wire,

$$E = \frac{\lambda}{2\pi\epsilon_0 x}$$

Is there an easier way to work out the electric field of an infinitely long line of charge?

Giancoli pg 592 Example 22-1

Calculate the electric flux through the rectangle shown. The rectangle is 10 cm by 20 cm, the electric field is uniform at 200 N/C, and the angle θ is 30° .



Non-uniform field and/or flux through curved surfaces

In general, when the field is not uniform and/or if the surface is not flat, we have to divide up the chosen surface into n smaller patches of area $\Delta A_1, \Delta A_2, \dots$ small enough that

- 1) it can be considered (nearly) flat
- 2) the variation of the field over the selected area is (nearly) constant.

Thus, the entire (electric) flux Φ_E over the surface is

$$\Phi_E \approx \sum_{i=1}^n \vec{E}_i \cdot \Delta \vec{A}_i$$

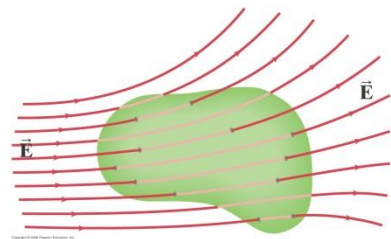
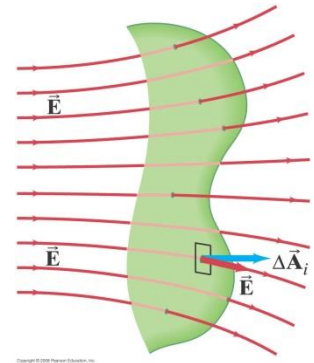
where $\vec{E}$ is the field passing through area $\Delta \vec{A}_i$. In the limit, where the elemental area is infinitesimally small $\Delta \vec{A}_i \rightarrow 0$, the $\vec{E}$ is constant and we have to move from a discrete sum to continuous sum

$$\Phi_E = \int \vec{E} \cdot d\vec{A}.$$

Later, we will be dealing with total flux through closed surface – a surface of any shape that completely encloses a volume of space, we write

$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

Here, the symbol $\oint$ denotes an integral over a closed surface.



Revising the Concept of Flux



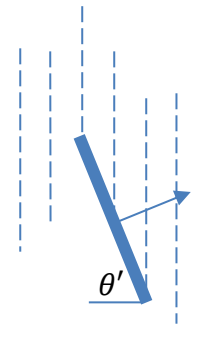
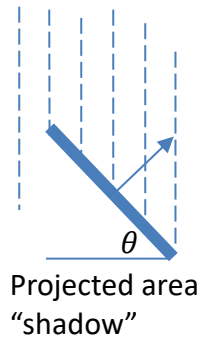
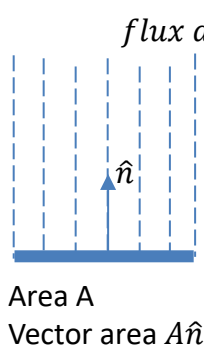
Heavy Rain – “high rain flux per unit area”
(high flux density)



Light Rain – “low rain flux per unit area”
(low flux density)

<http://99fm.com.na/unseasonably-heavy-rain-expected/>

<http://www.chachacharming.com/music/i-like-london-in-the-rain-mix/>



HOSY@ntu.edu.sg

Total flux “captured” by area $\Phi = \text{“flux density”} \times \text{Projected Area}$
 $= \text{“flux density”} \times \text{Area } \cos \theta$
 $= A |\vec{f}| |\hat{n}| \cos \theta = \vec{f} \cdot (A\hat{n})$

Earlier discussion on
gravitational fields:

For curved surface, $d\vec{A} = A d\hat{n}$

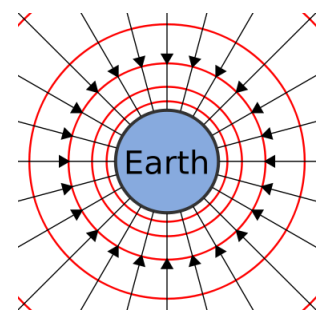
$$\text{Total flux, } \Phi = \int \vec{f} \cdot d\vec{A}$$

For closed curved surface, (convention, outwards +ve)

$$\text{Total flux, } \Phi = \oint \vec{f} \cdot d\vec{A}$$

Thus, for electric fields, the total electric flux Φ_E
passing through an imaginary surface
 (“Gaussian surface”) is given by

$$\Phi_E = \oint \vec{E} \cdot d\vec{A}$$

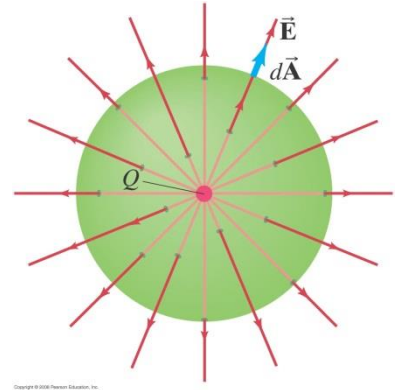


http://commons.wikimedia.org/wiki/File:Gravitational_field_Earth_lines_equipotentials.svg

$$\Phi_G = \oint \vec{g} \cdot d\vec{A}$$

A simple example

This all can look complicated later but let us try it on something not too difficult first. Consider a charge $Q = 1.2\text{nC}$ enclosed by an imaginary sphere of radius 10 cm with the charge in the centre of the sphere. Calculate the total electric flux that goes through the surface of the sphere. [Note here that electric field the magnitude of the electric field is constant on the surface and the elemental area $d\vec{A}$ is parallel to the $\vec{E}$ at all points on the sphere.]



[Convention: Flux leaving enclosed volume is +ve

Flux entering enclosed volume is -ve. This convention is consistent with E-field pointing away from +ve charge and towards -ve charge.]

What is the total electric flux if a large sphere of 20 cm is used?

What is the total electric flux if Q is replaced with a -ve charge with the same magnitude?

How will the total electric flux change if the charge is placed in the centre of a cube of sides 6 cm instead?

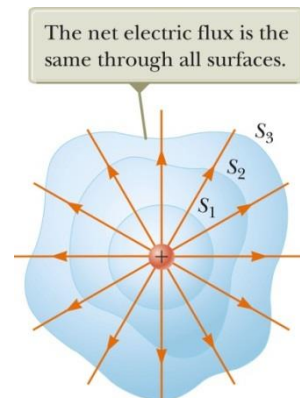
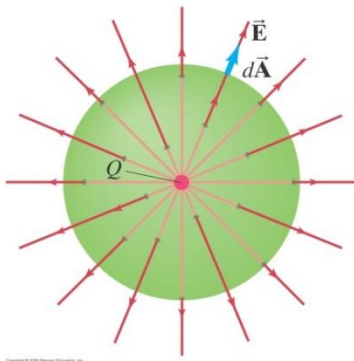
How will the total electric flux change if the charge is not in the center of sphere or the cube?

Giancoli pg 593, 22-2 Gauss's Law

The precise relationship between the electric flux through a closed surface (Gaussian surface) and the net charge enclosed within that surface is given by **Gauss's Law**:

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon}$$

where ϵ is permittivity as mentioned previously. In principle, Gauss's law can be used to find the expression for $\vec{E}$ for a system of charge or charge distribution. However, in practice, the application of Gauss's law is only limited to a number of highly symmetric electric fields. On the other hand, Gauss's law gives us a simple perception so that we can deal easily with situations such as when there are charges within the interior of a hollow sphere etc.



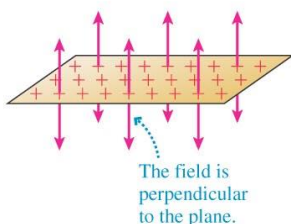
Why was that point charge / sphere example simple?

1. The magnitude of the electric field is constant on the surface and
2. the elemental area $d\vec{A}$ is parallel to the $\vec{E}$ at all points on the surface.

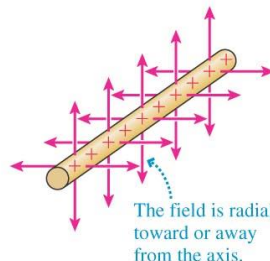
What are the other simple examples?

1. Point (Charge)
2. (Very long) Line (Charge per unit length, λ)
3. (Very Wide) Plane (Charge per unit area, σ)
4. 3D Sphere (Charge per unit volume, ρ)

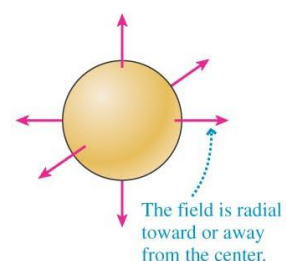
Planar symmetry



Cylindrical symmetry

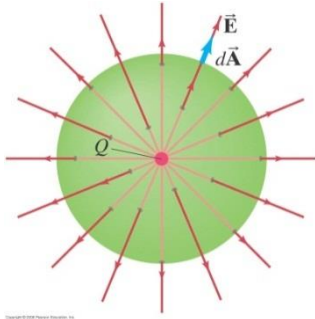


Spherical symmetry



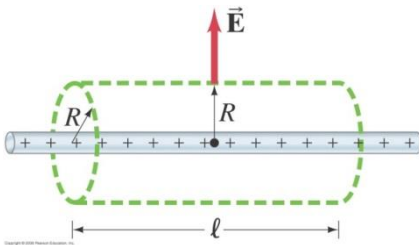
Basic symmetry:

Symmetry and regions of constant electric field



Total flux through surface of
'imaginary' sphere of radius $r = 4\pi r^2 E_{pt}$
Total charge enclosed = Q

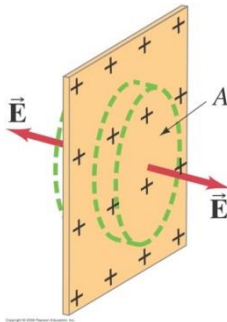
Applying Gauss's Law,



Wire with uniform charge per unit length = λ

Total flux through surface of 'imaginary' cylinder
of radius r and length $l = 2\pi R l E_L$
Total charge enclosed = λl

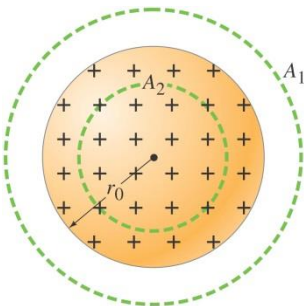
Applying Gauss's Law,



Conducting Plate with charge per unit area = σ

Total flux through surface of 'imaginary' disc
of cross sectional area $A = 2E_{pl}A$
Total charge enclosed = σA

Applying Gauss's Law,



Solid sphere of insulator with uniform charge density = ρ

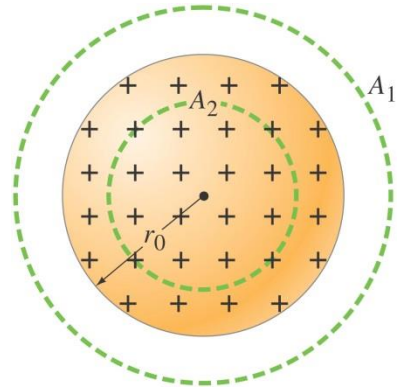
Total flux through surface of 'imaginary' sphere with
radius $r_1 (> r_0) = 4\pi r_1^2 E_{sp,ext}$
Total charge enclosed = $\rho \left(\frac{4}{3} \pi r_0^3 \right) = Q$

Applying Gauss's Law,

Spherically Symmetric Charge Distribution [Giancoli pg 596 Example 22-4]

An electric charge Q is distributed uniformly throughout a nonconducting sphere of radius r_0 . Determine the electric field

- (a) outside the sphere ($r > r_0$) and
- (b) inside the sphere ($r < r_0$).



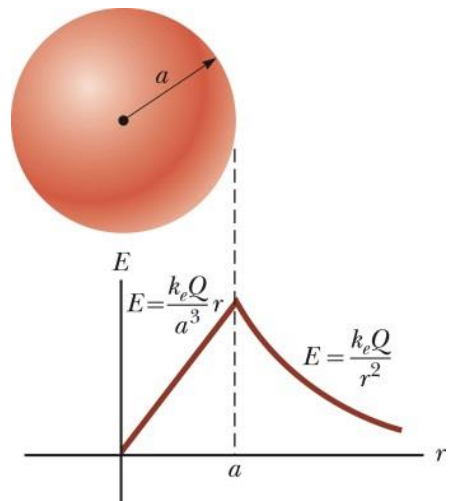
Mathematical concept test:

A sphere has mass M . What is the mass of a sphere made of the same material but double the radius?

- b) Total flux (outwards) through Gaussian surface $A_2 =$ _____

Total charge enclosed by Gaussian surface $A_2 =$ _____

Applying Gauss's Law:



c.f Notes on gravitation – what is the g-field inside Earth.

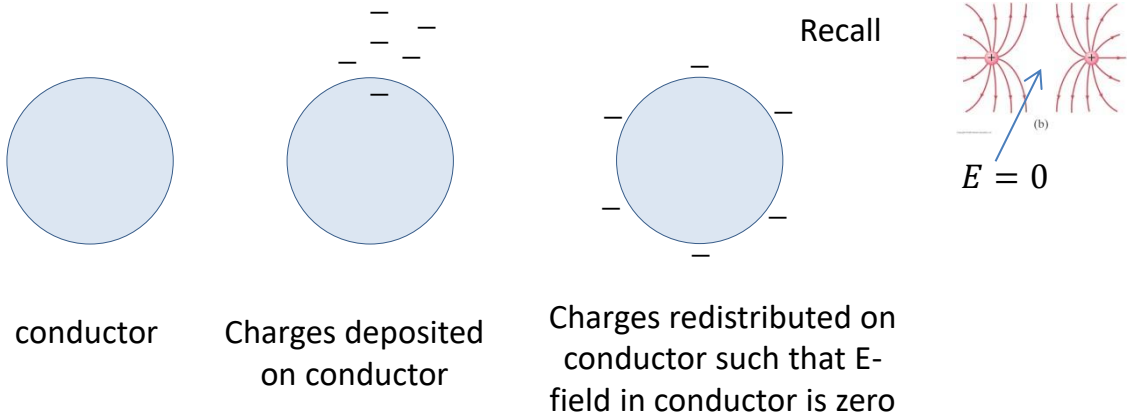
Electric Field inside conductor at Electrostatic Equilibrium I

[Giancoli pg 577, 598-600]

Electrostatic equilibrium – The electric field inside a conductor is zero when the conductor is in electrostatic equilibrium - charges are at rest. If there is a non-zero electric field within the conductor, there would be a force on the free electrons. The electrons would move until they reached positions where the electric field, hence electric force on them, become zero.

A few consequences at electrostatic equilibrium:

1. The electric field is zero everywhere inside the conductor (hollow or solid).
2. If conductor is isolated, any net charge will distribute itself on the surface.

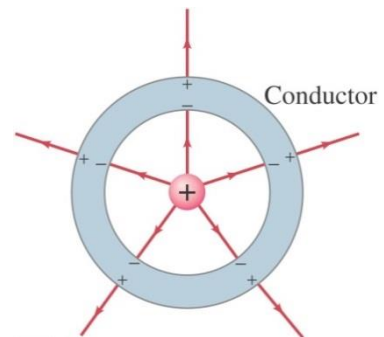


Google homework: Faraday cage.

3. If a charge (say $+Q$) is surrounded by an isolated uncharged conducting spherical shell, applying Gauss's law with the Gaussian surface inside the conductor,

We can infer that there is

Since the shell is neutral,



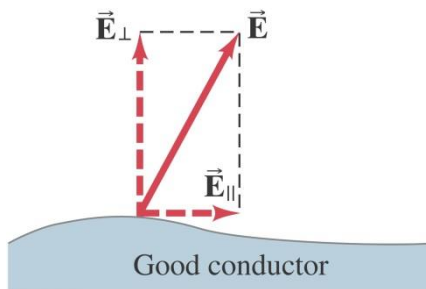
Though no field exists inside conductor, an E-field exists outside of it, as if the conducting spherical shell were not there.

Electric Field inside conductor at Electrostatic Equilibrium II

[Giancoli pg 577, 598-600]

A few consequences at electrostatic equilibrium:

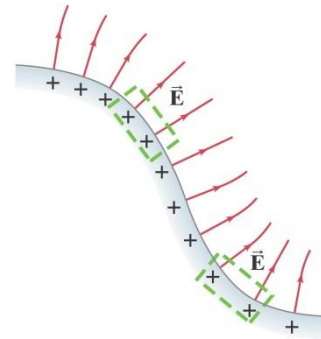
- The electric field near the surface of the conductor points perpendicular to the surface. If the electric field $\vec{E}$ at the surface of a conductor had a component parallel to the surface $\vec{E}_{||}$ it would exert a force on the electrons at the surface causing electrons to move along surface until the net force on the system of free charges become zero. Only the component of the electric field perpendicular to the surface $\vec{E}_{\perp}$ will not vanish.



No in electrostatic equilibrium

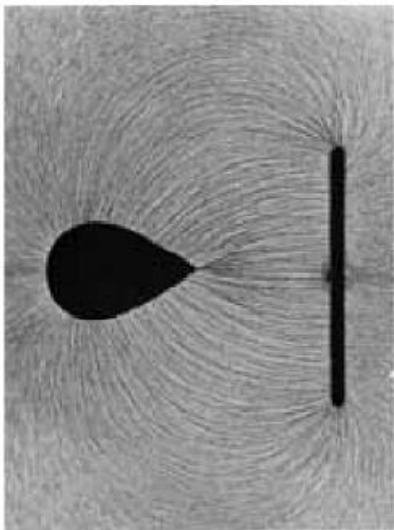
$$\vec{E}_{\perp} \neq 0$$

$$\vec{E}_{||} = 0$$



In electrostatic equilibrium

- For irregularly shaped conductor, the surface density is greatest at locations where the radius of curvature of the surface is smallest.



Electric field pattern of a charged conducting plate placed near an oppositely charged pointed conductor. Small pieces of thread suspended in oil align with the electric field lines. The field surrounding the pointed conductor is most intense near the pointed end and at other places where the radius of curvature is small.

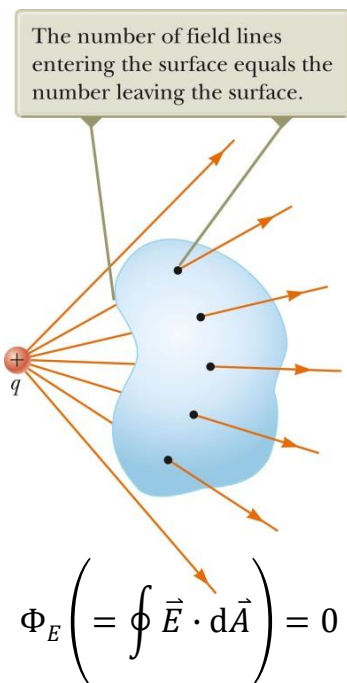
<http://www.kshitij-iitjee.com/electric-potential-due-to-charged-conductor>

Giancoli pg 577 Conceptual Example 21-14: Shielding, and safety in a storm.

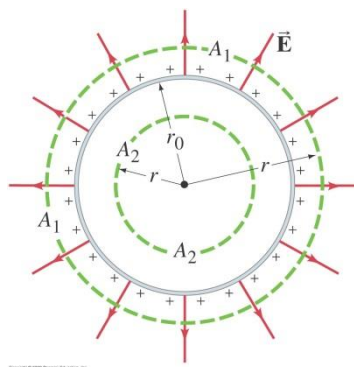
A neutral hollow metal box is placed between two parallel charged plates as shown. What is the field like inside the box?



Three possible cases whereby $\Phi_E = 0$



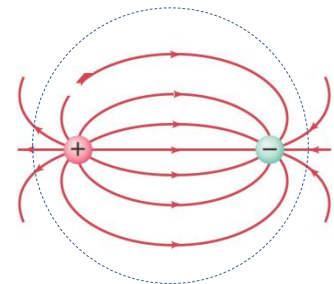
and here E
Net charge enclosed =



For A_2

$$\Phi_E \left(= \oint \vec{E} \cdot d\vec{A} \right) = 0$$

and here E
Net charge enclosed =

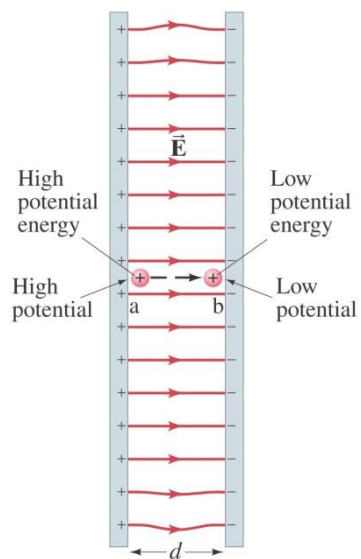
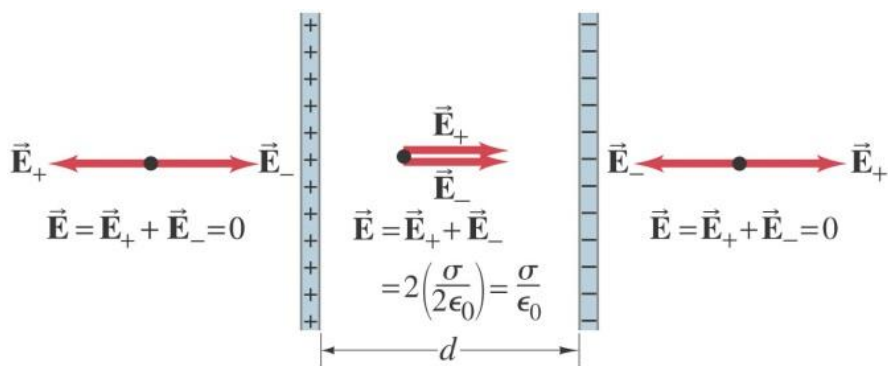


Here,

$$\Phi_E \left(= \oint \vec{E} \cdot d\vec{A} \right) =$$

and here E
Net charge enclosed =

Electric Field of Parallel plates



From Earlier

2026S1 PH1012: Physics A

Can you produce a magnetic field without a magnet? – Ampere's Law

Dr Ho Shen Yong

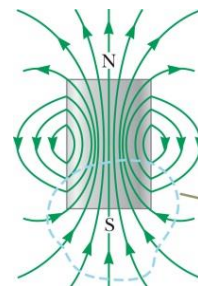
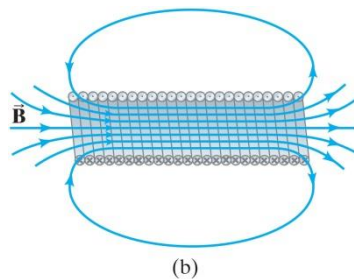
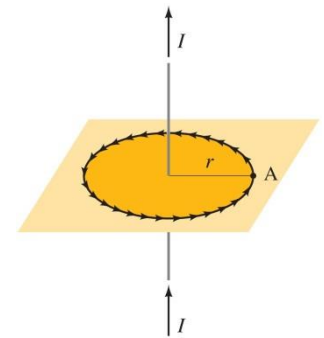
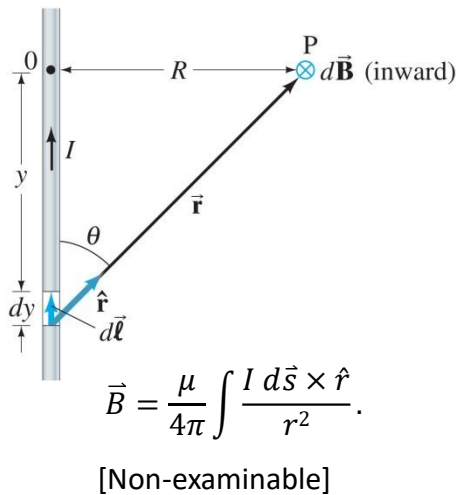
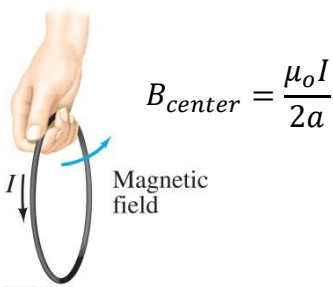
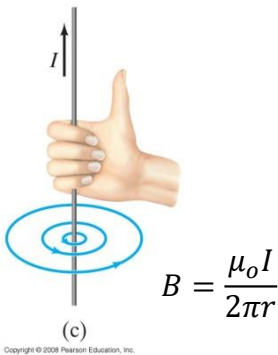
Lecturer, School of Physical and Mathematical Sciences
Nanyang Technological University

Week 11

Serway Chap 29.1, 29.3 – 29.4

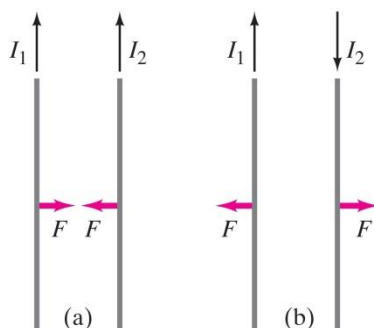
"Tesla has contributed more to electrical science than any man up to his time."

- Statement of 1896, Lord William Thomson Kelvin (1824-1907)



The net magnetic flux through a closed surface surrounding one of the poles or any other closed surface is zero.

$$\oint \vec{B} \cdot d\vec{A} = 0$$



Magnetic Fields

Giancoli pg 736 Definitions of the Ampere and Coulomb

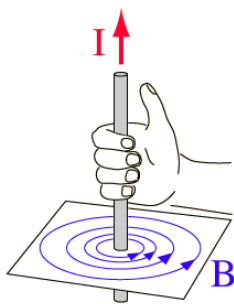
The ampere is officially defined in terms of the force between two current-carrying wires:

One ampere is defined as that current flowing in each of two long parallel wires 1 m apart, which results in a force of exactly 2×10^{-7} N per meter of length of each wire.

$$\frac{F}{l} = \frac{\mu_o I_1 I_2}{2\pi d}$$

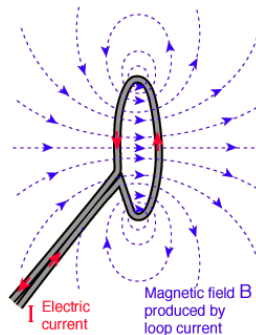
The coulomb is then defined as exactly one ampere-second.

Where did the formulas for magnetic fields come from?



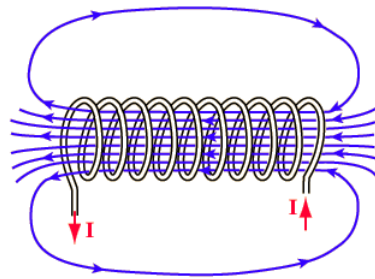
Long Wire

$$B = \frac{\mu_o I}{2\pi r}$$



Flat Coil

$$B_{center} = \frac{\mu_o I}{2a}$$



Long Solenoid

$$B_{center} = \mu_o n I$$

Electric Field	Magnetic field
Coulomb's Law $E = \frac{1}{4\pi\epsilon_o} \frac{Q}{r^2}$	Biot-Savart's Law $\vec{B} = \frac{\mu q}{4\pi} \frac{\vec{v} \times \hat{r}}{r^2}$
Gauss's Law $\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{net, encl}}{\epsilon_o}$	Ampere's Law $\oint \vec{B} \cdot d\vec{l} = \mu_o I_{net, encl}$

Biot-Savart Law [Knight pg 1003-1005; Giancoli pg 743-745]

[Non-examinable] Video after Exam

As we recall, we can use the right hand grip to predict the direction of a magnetic field from current. Thus, not surprisingly, we will need to use our knowledge of cross product.

The starting point is the idea that moving charges are the source of magnetic fields. The diagram on the right shows a charged particle q moving with velocity $\vec{v}$. The magnetic field at a point $\vec{r}$ (with the charge as the origin) is given by

$$\vec{B} = \frac{\mu q v \sin \theta}{4\pi r^2}.$$

This is known as Biot-Savart's law for point charges. The inverse squared relation resembles that of Coulomb's law and Newton's law of gravitation. We will explain more about the quantity known as permeability μ later. In vector form, we can write

$$\vec{B} = \frac{\mu q}{4\pi} \frac{\vec{v} \times \hat{r}}{r^2}$$

where $\hat{r}$ is the unit vector for $\vec{r}$. Considering a small amount of charge ΔQ a small length Δs with velocity $\vec{v} = \frac{\Delta s}{\Delta t}$, we can write

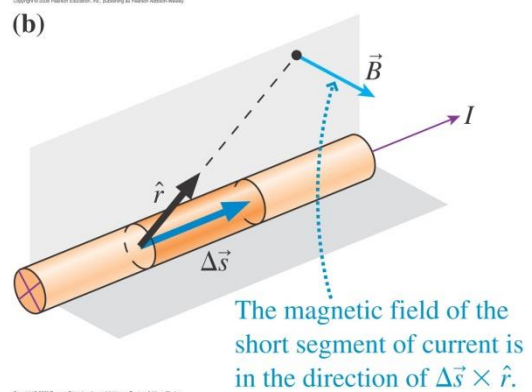
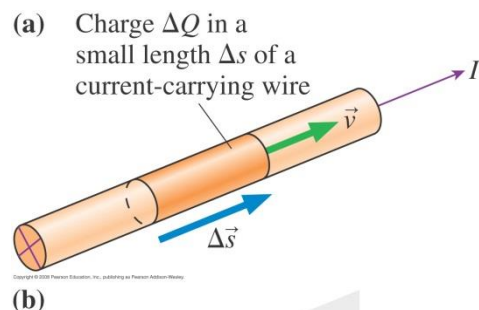
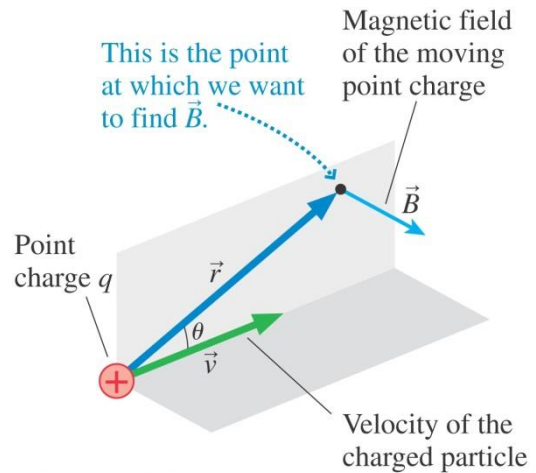
$$(\Delta Q)\vec{v} = \Delta Q \frac{\Delta s}{\Delta t} = \frac{\Delta Q}{\Delta t} \Delta s = I \Delta s$$

Replacing $q\vec{v}$ with $I d\vec{s}$, we get

$$d\vec{B} = \frac{\mu}{4\pi} \frac{I d\vec{s} \times \hat{r}}{r^2}$$

Thus, for the magnetic field at $\vec{r}$

$$\vec{B} = \frac{\mu}{4\pi} \int \frac{I d\vec{s} \times \hat{r}}{r^2}.$$



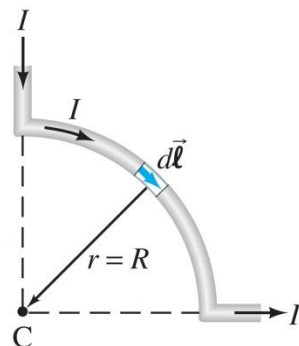
Giancoli pg 745 Example 28.13 [Non-examinable] Video after Exam

One quarter of a circular loop of wire carries a current I . The current I enters and leaves on straight segments of wire, as shown; the straight wires are along the radial direction from the center C of the circular portion. Find the magnetic field at point C .

$$\vec{B} = \frac{\mu}{4\pi} \int \frac{I d\vec{s} \times \hat{r}}{r^2}.$$

$$dB = \frac{\mu_o I dl}{4\pi r^2}$$

$$B = \int dB = \int \frac{\mu_o I dl}{4\pi r^2} = \frac{\mu_o I}{4\pi r^2} \int dl = \frac{\mu_o I}{4\pi r^2} \left(\frac{1}{4} \times 2\pi R \right) = \frac{\mu_o I}{8R}$$



Giancoli pg 744 Example 28.11 [Non-examinable]

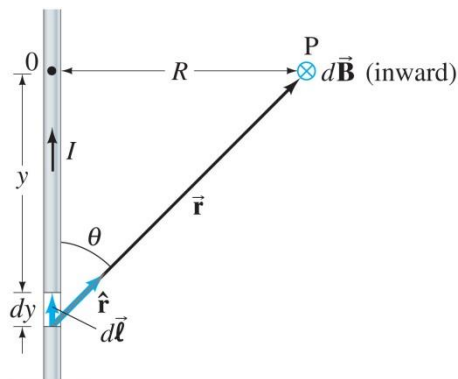
For the field near a long straight wire carrying a current I , show that the Biot-Savart law gives

$$B = \frac{\mu_o I}{2\pi r}.$$

$$B = \frac{\mu_o I}{2\pi} \int_{y=-\infty}^{\infty} \frac{dy \sin \theta}{r^2} = \dots$$

...

$$B = \frac{\mu_o I}{2\pi R} \int_{\theta=0}^{\pi} \sin \theta d\theta = \dots = \frac{\mu_o I}{2\pi R}.$$



Ampere's Law

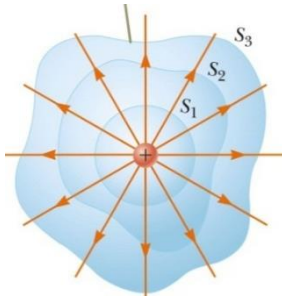
As we have seen previously in Gauss's Law, by exploiting the high degree of symmetry for certain charge distributions, we have very simple ways of obtaining the E-fields for these distributions. There is an analogous theorem for current and magnetism – Ampere's Law.

E-fields: Coulomb's Law (Static charges) → Gauss's Law (closed surface)

B-fields: Biot-Savart's Law (Moving charges) → Ampere's Law (closed path/line)

In Gauss's law, we think about $\vec{E} \cdot d\vec{A}$ - field lines crossing areas. In Ampere's law, we will think about $\vec{B} \cdot d\vec{s}$ - dot product of field lines along a path.

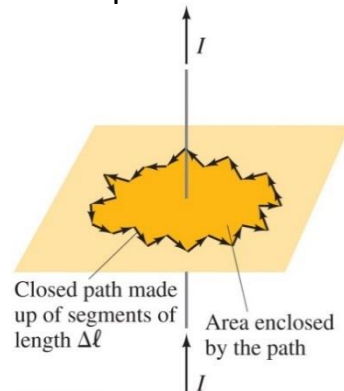
Gauss' s Law



$$\sum E_{\perp} \Delta A = \frac{Q_{net, encl}}{\epsilon_0}$$

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{net, encl}}{\epsilon_0}$$

Ampere' s Law



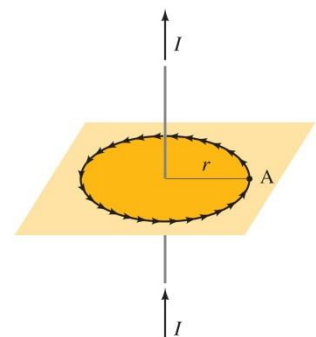
$$\sum B_{\parallel} \Delta l = \mu_o I_{net, encl}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_o I_{net, encl}$$

Ampere's Law: The line integral of $\vec{B} \cdot d\vec{s}$ around any closed path equals μI , where I is the total steady current passing through any surface bounded by the closed path:

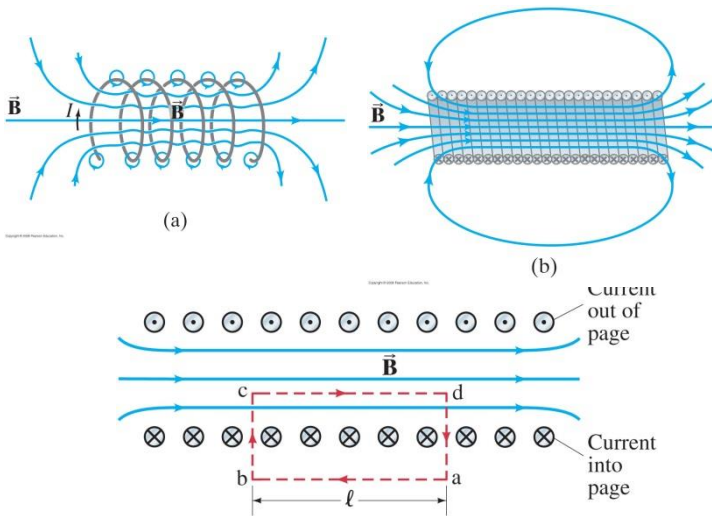
$$\oint \vec{B} \cdot d\vec{l} = \mu_o I_{net, encl}.$$

To illustrate, how easy things have become for certain currents, let us revisit the derivation of the B-field for the straight wire.



Giancoli pg 741 Magnetic field of a long tightly wound Solenoid

A solenoid is a coil of wire containing many loops. To find the field inside, we use Ampère's law along the path indicated in the figure.



The magnetic field at the side of the solenoid is approximately half of that in the middle.

$$|\vec{B}_{end}| \approx \frac{1}{2} |\vec{B}_{center}|$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{net, encl.}$$

$$B_{ab}l_{ab} + B_{bc}l_{bc} + B_{cd}l_{cd} + B_{da}l_{da} = \mu_0 (\quad)$$

Gauss's Law in magnetism

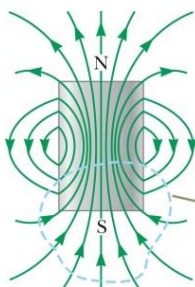
Earlier, we saw Gauss's law for electric fields:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon}$$

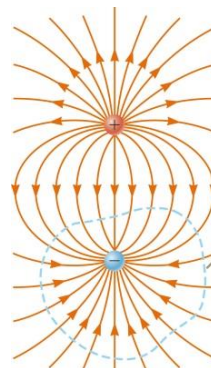
The electric flux in an enclosed area is proportional to the total charge the area encloses. A similar question can be asked of magnetic fields

$$\oint \vec{B} \cdot d\vec{A} = ?$$

The answer is _____. Unlike electric fields, magnetic field lines have no 'ends'. This statement represents that isolated magnetic monopoles has never been detected and may never exist.



The net magnetic flux through a closed surface surrounding one of the poles or any other closed surface is zero.



The electric flux through a closed surface surrounding one of the charges is not zero.

Faraday's Law (Again): Induced e.m.f. s and electric fields

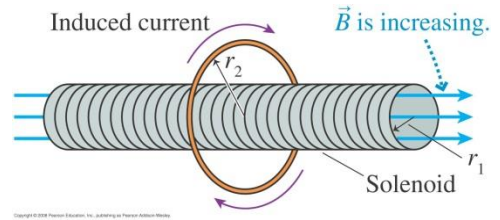
[Giancoli 29.7 pg 773] [Non-Examinable] EEE Video after Exam

Consider a long, tightly wound solenoid of radius r_1 passing through a larger conducting loop of radius r_2 . Even though the loop is completely outside the magnetic field of the solenoid, the changing magnetic field in the solenoid still causes an induced current in the loop. How does the charge carriers in loop 'know' about the changing B-field?

What happens here is that the changing magnetic flux not only induces an e.m.f. in the loop but also a whole region of electric field. Thus, there are two ways to produce an electric field: i) by charges and by ii) a changing magnetic field! Similar to what we have seen in Ampere's law, a line integral round a closed path of electric field encloses a changing magnetic flux can be written

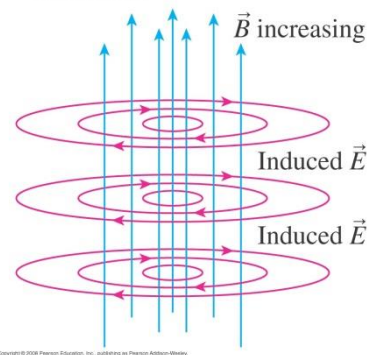
$$\mathcal{E} = \oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

Electric fields due to changing magnetic fields are non-conservative since $\oint \vec{E} \cdot d\vec{s} \neq 0$.



[Knight pg 1058, Ex 34.31]

(b) The induced electric field circulates around the magnetic field lines.



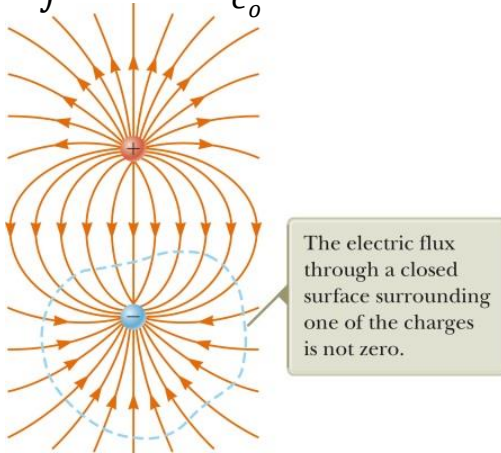
[Knight pg 1060, Ex 34.34b]

Maxwell's Equation

We will do a short summary of the integral equations that we have encountered so far. In fact, we have seen all four of Maxwell's equations.

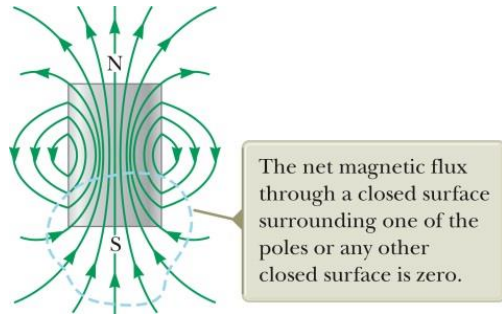
Coulomb's Law

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$



No Magnetic Monopoles

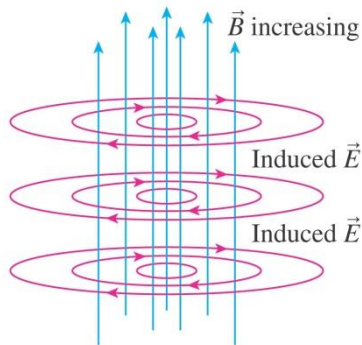
$$\oint \vec{B} \cdot d\vec{A} = 0$$



Faraday's Law

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$$

(b) The induced electric field circulates around the magnetic field lines.



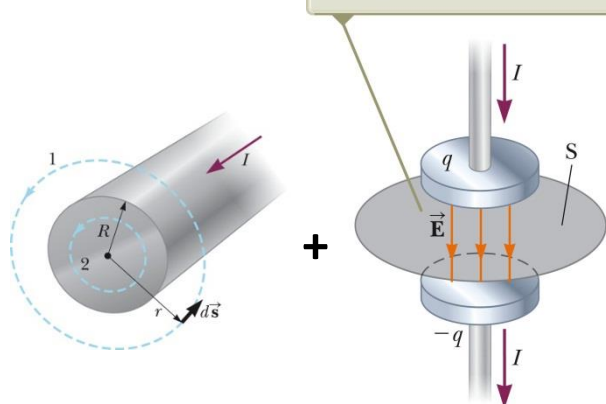
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Changing B-field -> E-field.

Ampere-Maxwell Law

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

The electric field lines between the plates create an electric flux through surface S.



Changing E-field -> B-field.

$$c = 1/\sqrt{\mu_0 \epsilon_0}$$

Conventions and Mnemonics

Figure 1: Current and Magnetic Field

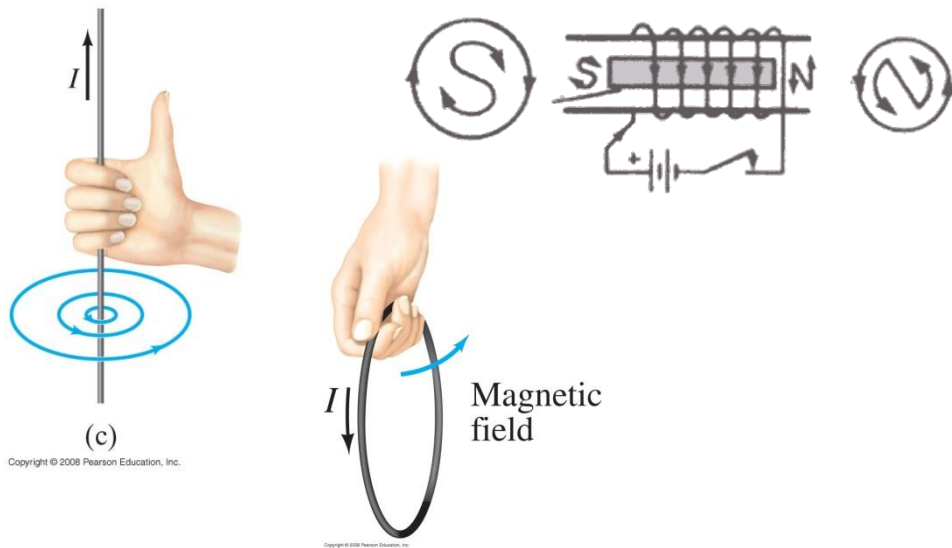


Figure 2: Current in Magnetic Field

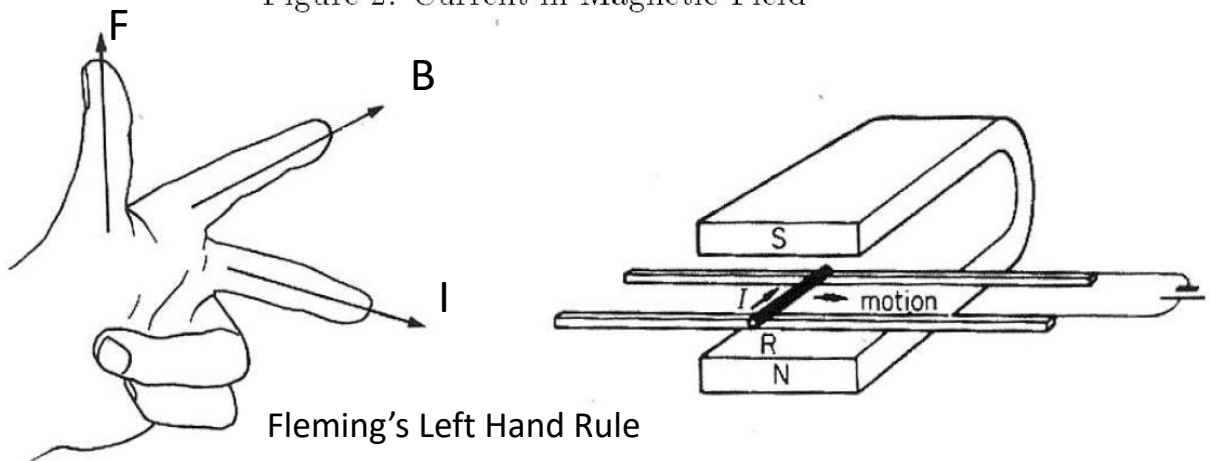


Figure 3: Induced E.M.F

